BITS WILP / NSP1

# EvalSphere

Benchmark Smarter. Decide Faster. · Assignment 2

## 01 GITHUB REPOSITORY

Complete source code · requirements.txt · README.md · test CSV data

https://github.com/sreeram348/2026ak05608_assignment_2.git

## 02 LIVE STREAMLIT COMMUNITY CLOUD APP

https://2026ak05608assignment2git-tgcyacyp3j2ftfuwllshbx.streamlit.app/

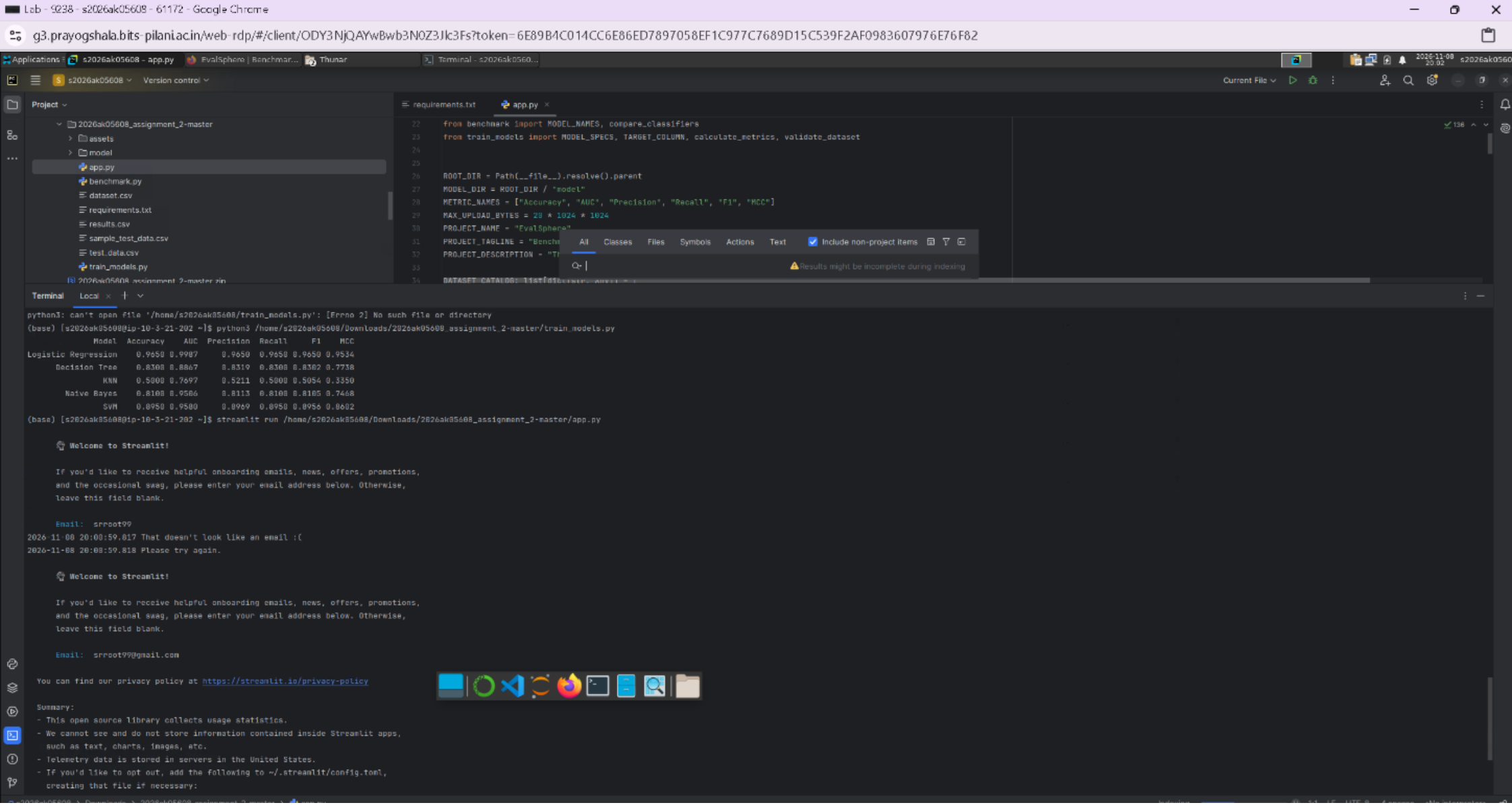
## 03 LAB EXECUTION & APPLICATION SCREENSHOTS

Four screenshots follow: model training results, the running Streamlit server, the assignment workspace, and the live app open in the lab browser.

**Created by SRIRAM POLAMARASETTI**

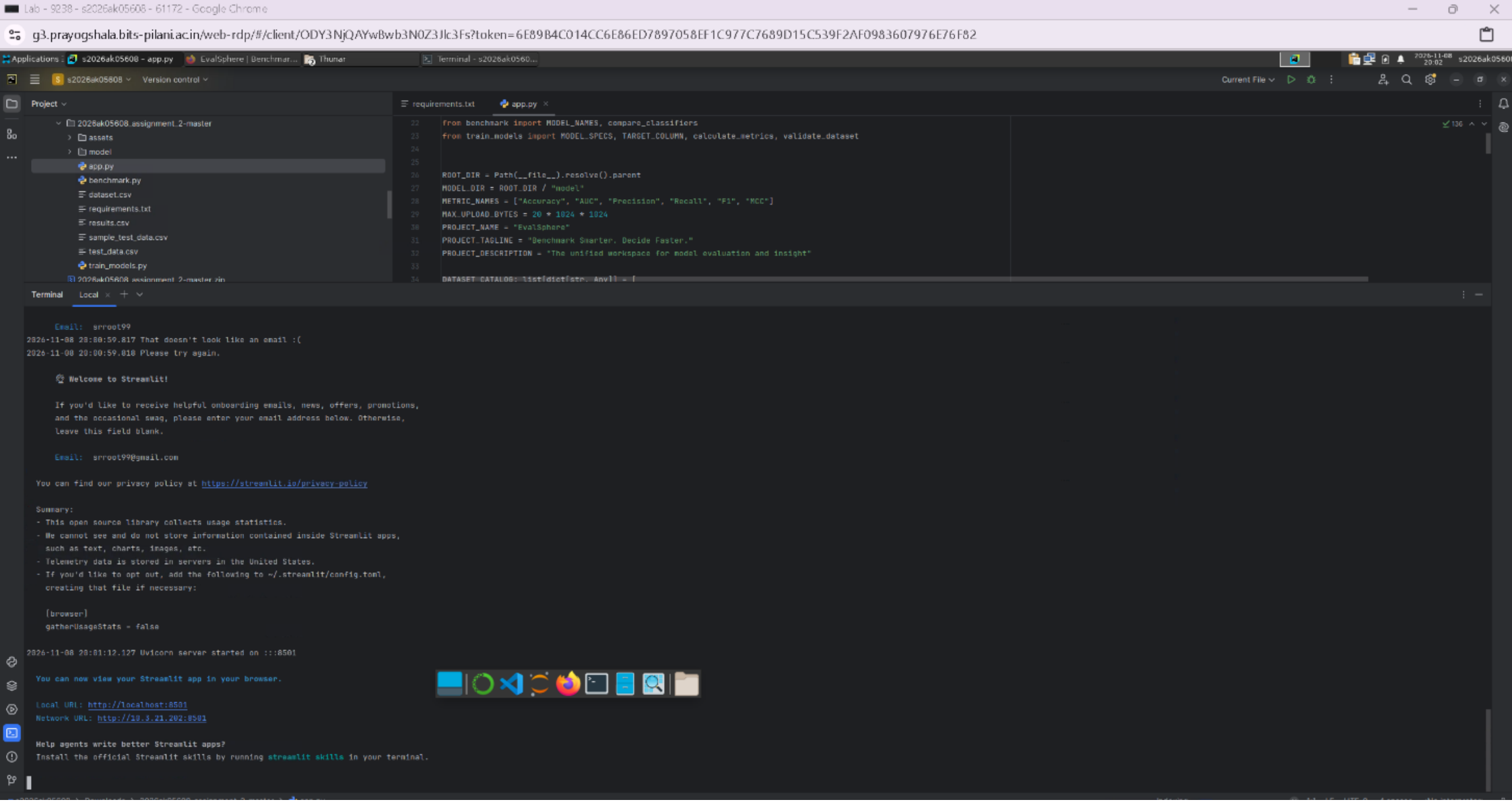
BITS ID 2026ak05608

Project details and user instructions follow the screenshots.



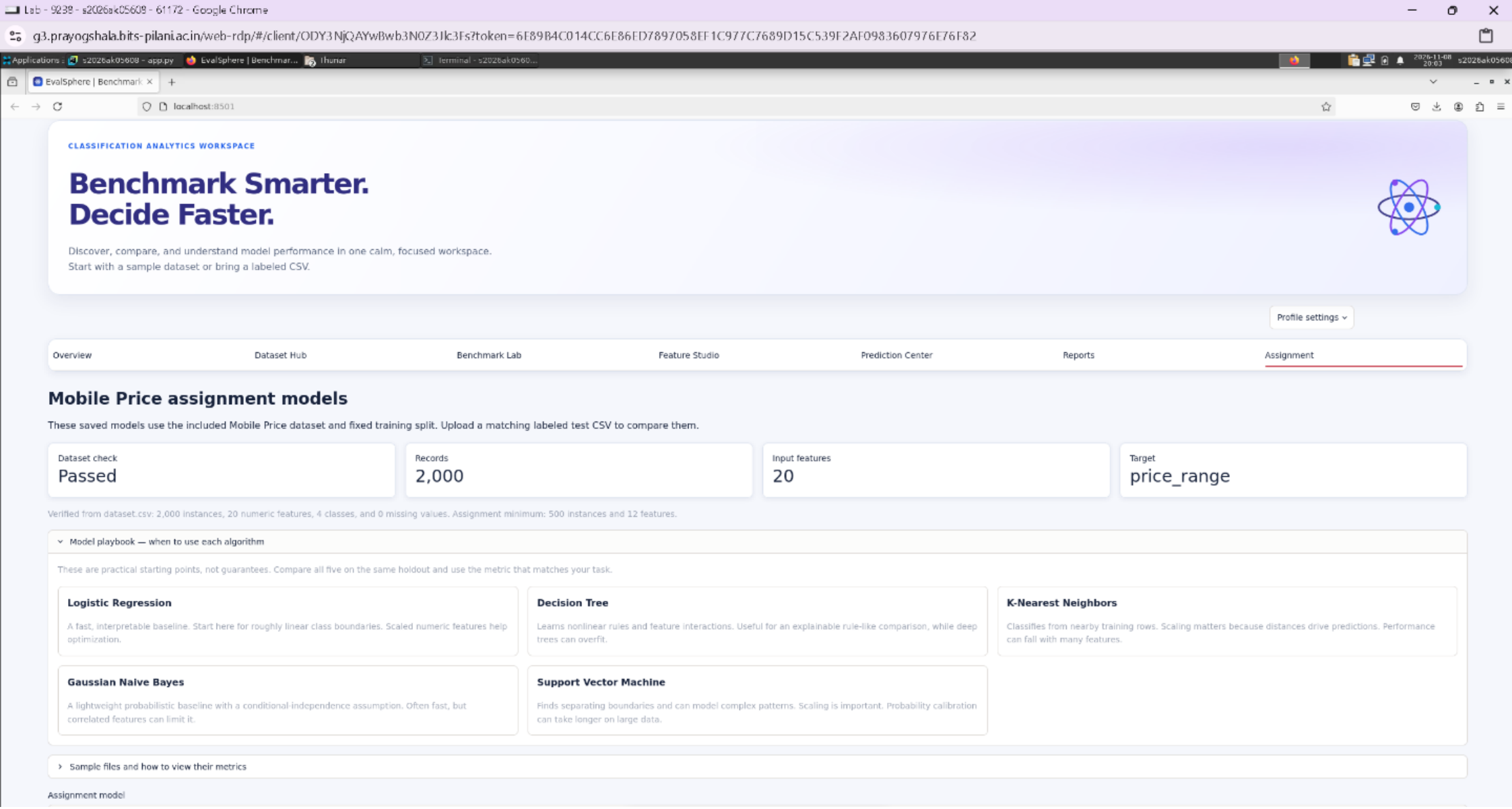

## 01 Model training and benchmark results

The lab workspace with the project code and the five-classifier evaluation output.



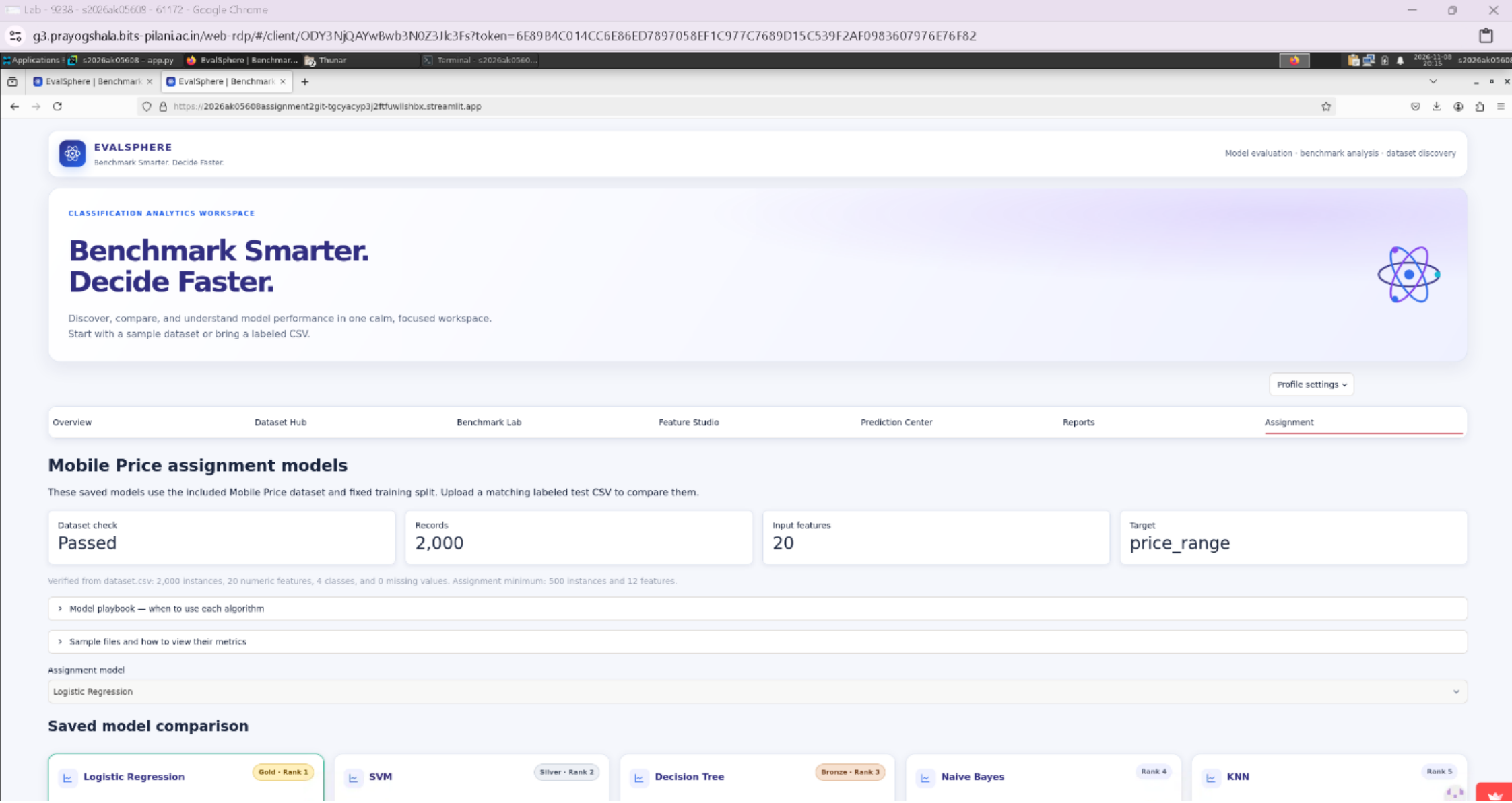

## 02 Streamlit application launch

The lab terminal showing the Streamlit server running and the local application URL.




## 03 EvalSphere assignment workspace

The application displaying the Mobile Price assignment and saved-model comparison workflow.




## 04 Live Streamlit app in the BITS Virtual Lab

The deployed EvalSphere app open in the lab browser, showing the assignment results workspace.

# EvalSphere

Benchmark Smarter. Decide Faster.

Student: SRIRAM POLAMARASETTI BITS ID: 2026ak05608

EvalSphere is a machine-learning classification and benchmarking workspace. It compares five classifiers, explores dataset characteristics, evaluates predictions, and presents the Mobile Price Classification assignment results.

## Project links

- Repository URL: Open the GitHub repository (https://github.com/sreeram348/2026ak05608_assignment_2.git)
- Live app URL: Open EvalSphere (https://2026ak05608assignment2git-tgcyacyp3j2ftfuwllshbx.streamlit.app/)

## Dataset

The assignment uses the Mobile Price Classification dataset.

| Property | Value |
| --- | --- |
| **Task** | Multiclass classification |
| **Records** | 2,000 |
| **Input features** | 20 numeric mobile specifications |
| **Target** | price_range (four classes: 0–3) |
| **Class distribution** | 500 records per class |
| **Missing values** | None |

The dataset exceeds the assignment requirements of 500 records and 12 input features. A stratified 80/20 split with random seed 42 is used. Preprocessing is fitted on the training partition only. The 400-row reference holdout and a balanced 100-row sample are included for evaluation.

# Models and results

The five classifiers are evaluated on the same holdout. Precision, recall, and F1 use weighted averaging; AUC uses weighted one-vs-rest.

| Model | Accuracy | AUC | Precision | Recall | F1 | MCC |
|---|---|---|---|---|---|---|
| **Logistic Regression** | 0.9650 | 0.9987 | 0.9650 | 0.9650 | 0.9650 | 0.9534 |
| **Decision Tree** | 0.8300 | 0.8867 | 0.8319 | 0.8300 | 0.8302 | 0.7738 |
| **K-Nearest Neighbors** | 0.5000 | 0.7697 | 0.5211 | 0.5000 | 0.5054 | 0.3350 |
| **Gaussian Naive Bayes** | 0.8100 | 0.9506 | 0.8113 | 0.8100 | 0.8105 | 0.7468 |
| **Support Vector Machine** | 0.8950 | 0.9580 | 0.8969 | 0.8950 | 0.8956 | 0.8602 |

Logistic Regression is the top performer on this fixed holdout, with 96.50% accuracy and the highest AUC, F1, and MCC. Results describe this evaluation split and do not guarantee performance on other data.

# Using the application

The app is organized into seven workspaces:

| Workspace | Purpose |
|---|---|
| **Overview** | Review the selected dataset and benchmark summary. |
| **Dataset Hub** | Search and load a sample dataset or upload a labeled CSV. |
| **Benchmark Lab** | Run and compare the five classifiers with interactive, labeled charts. |
| **Feature Studio** | Explore feature relevance, distributions, and correlations. |
| **Prediction Center** | Review holdout predictions and confidence, then export results. |

**Reports**

Purpose: Preview dataset and benchmark summaries and export CSV or PDF reports.

**Assignment**

Purpose: View dataset verification, saved-model comparisons, and the confusion matrix and classification report together.

## Run a benchmark

1. Open Dataset Hub and load a sample or upload a labeled classification CSV. 2. Select the target column and the holdout size. 3. Open Benchmark Lab and choose Run benchmark. 4. Compare the champion model, metric cards, leaderboard, and performance charts. 5. Explore features, review predictions, or export a report from the other workspaces.

## Review the assignment results

Open Assignment to see the Mobile Price dataset check, saved-model comparison, and confusion matrix for the bundled holdout. Select a model to review its classification report. Upload a matching labeled test CSV to evaluate the saved models without retraining.

# Run locally

Use Python 3.11 or newer.

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
python train_models.py
streamlit run app.py
```

train_models.py trains the saved models, creates the test CSV files, and updates results.csv.